

Software Atelier IV

Andrea Mocci

Software Institute - Università della Svizzera italiana

TAs: Stefano Campanella, De Vita Gianmarco,
Jamshidiyantehrani Masoud, Puccinelli Niccolò,
Lamberto Ragnolini

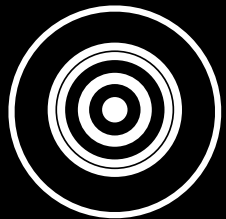
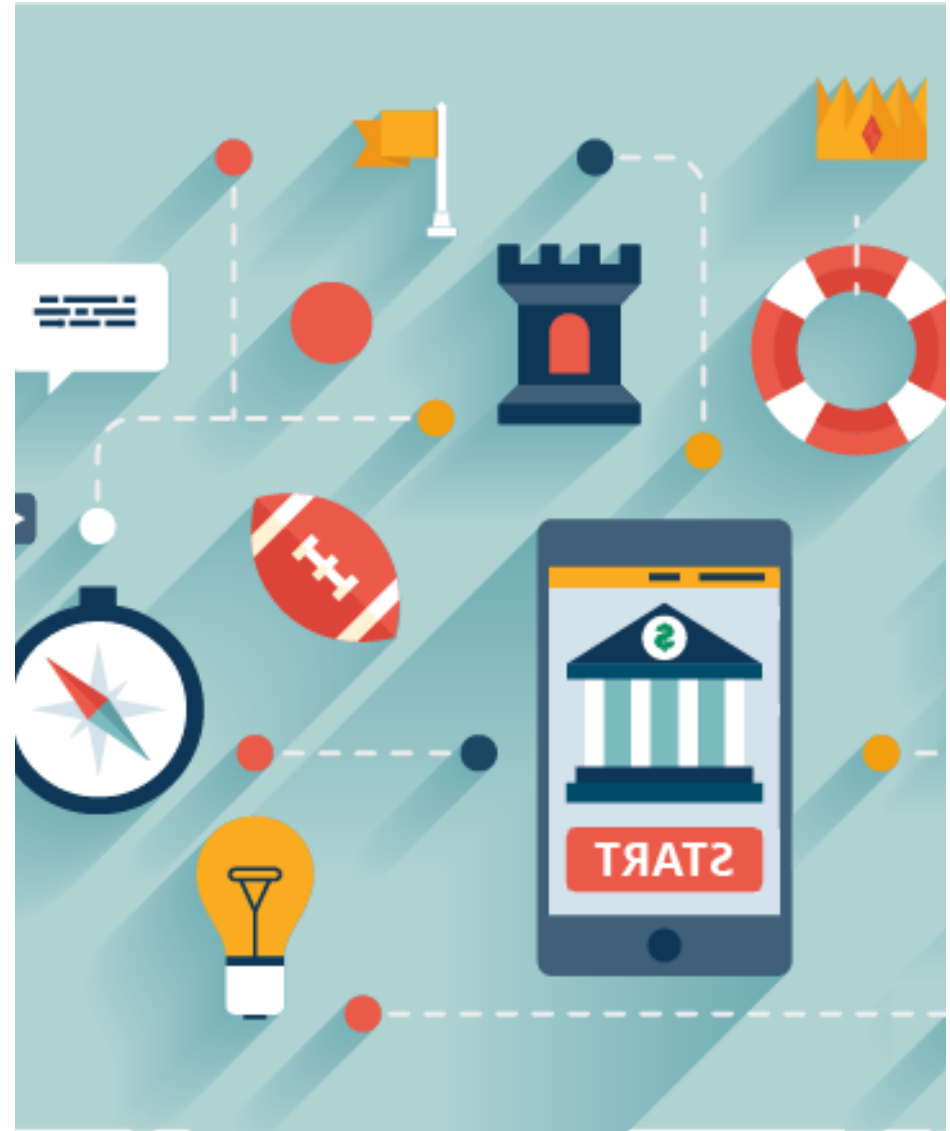
Software Institute, Università della Svizzera italiana



Project

SEWorld

A platform to learn (SA4)
Software Engineering
topics with the help of AI



Goal
of the Project



Eating your own dog food

 [7 languages](#) 

[Article](#) [Talk](#)

[Tools](#) 

From Wikipedia, the free encyclopedia

Eating your own dog food or "**dogfooding**" is the practice of using one's own products or services.^[1] This can be a way for an organization to test its products in real-world usage using [product management](#) techniques. Hence dogfooding can act as [quality control](#), and eventually a kind of [testimonial advertising](#). Once in the market, dogfooding can demonstrate developers' confidence in their own products.^{[2][3]}



Goal
of the Project

You will develop a platform where you can practice exercises similar to the ones you will find in the final - including topics like testing, RE, quality, and software architecture.



Goal
of the Project

We will see how integrate
an AI to support both the
teacher to create new
exercises and the students
to get feedback,
explanations, etc. - but in a
critical way



Goal
of the Project

We will introduce **minimal gamification elements** to represent the skills acquired by the students and the amount of exercises solved.



Goal
of the Project

Backend

Domain Model + Data Access



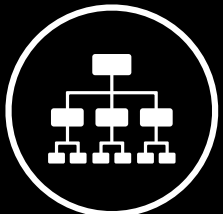
Will Start after Lab02
next week

Frontend

Presentation

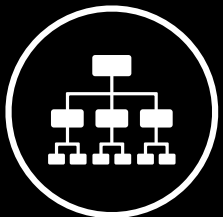
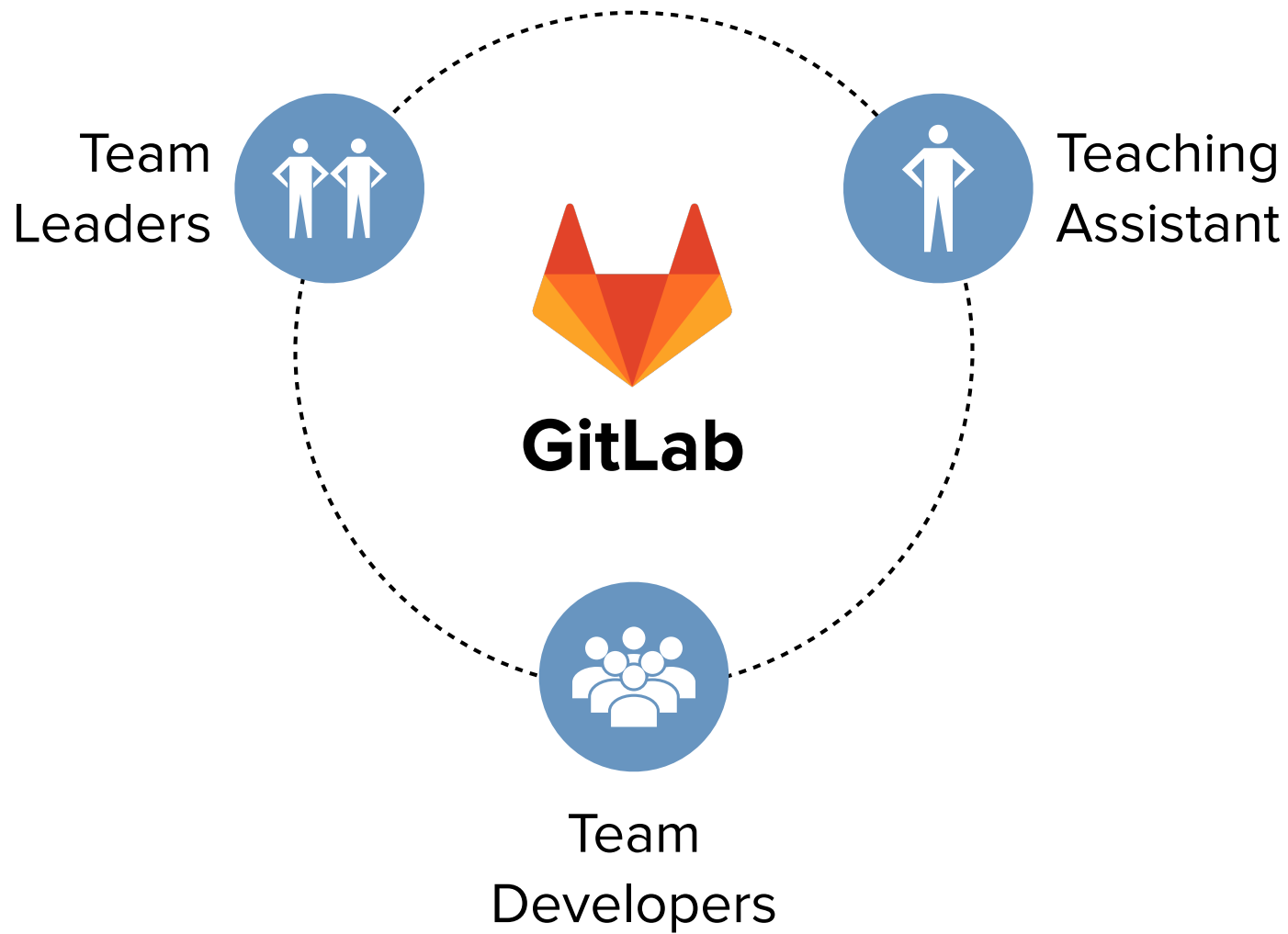


Will Start after Lab03
in 2 weeks



Project
Tech

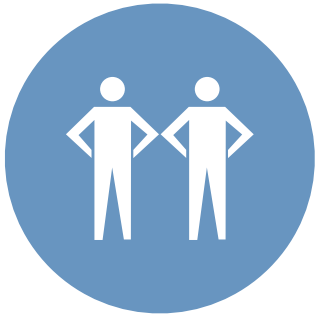
Team Roles



Project
Organization



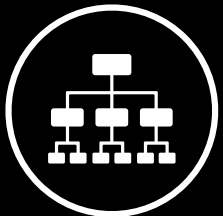
Teaching
Assistant



Team
Leaders



Team
Developers

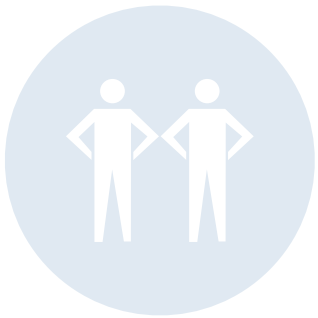


Project
Organization



**Teaching
Assistant**

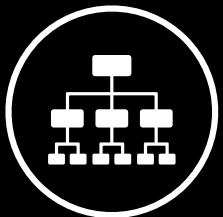
Any TA according to availability, or
the Instructor if needed



Act as Customer Representative /
Proxy



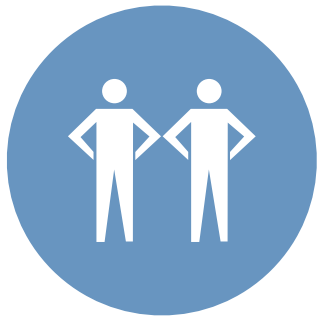
Help the team leaders on
coordination and planning



**Project
Organization**



Collaborate and coordinate team members

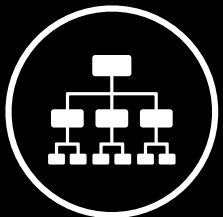


Team
Leaders

Responsible for team
accountability (i.e., assign issues to team members)



Responsible to **merge Merge Requests** from the team into the development and main branch



Project
Organization

Leaders Responsibilities

In addition to **development**:

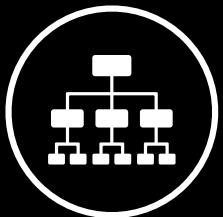
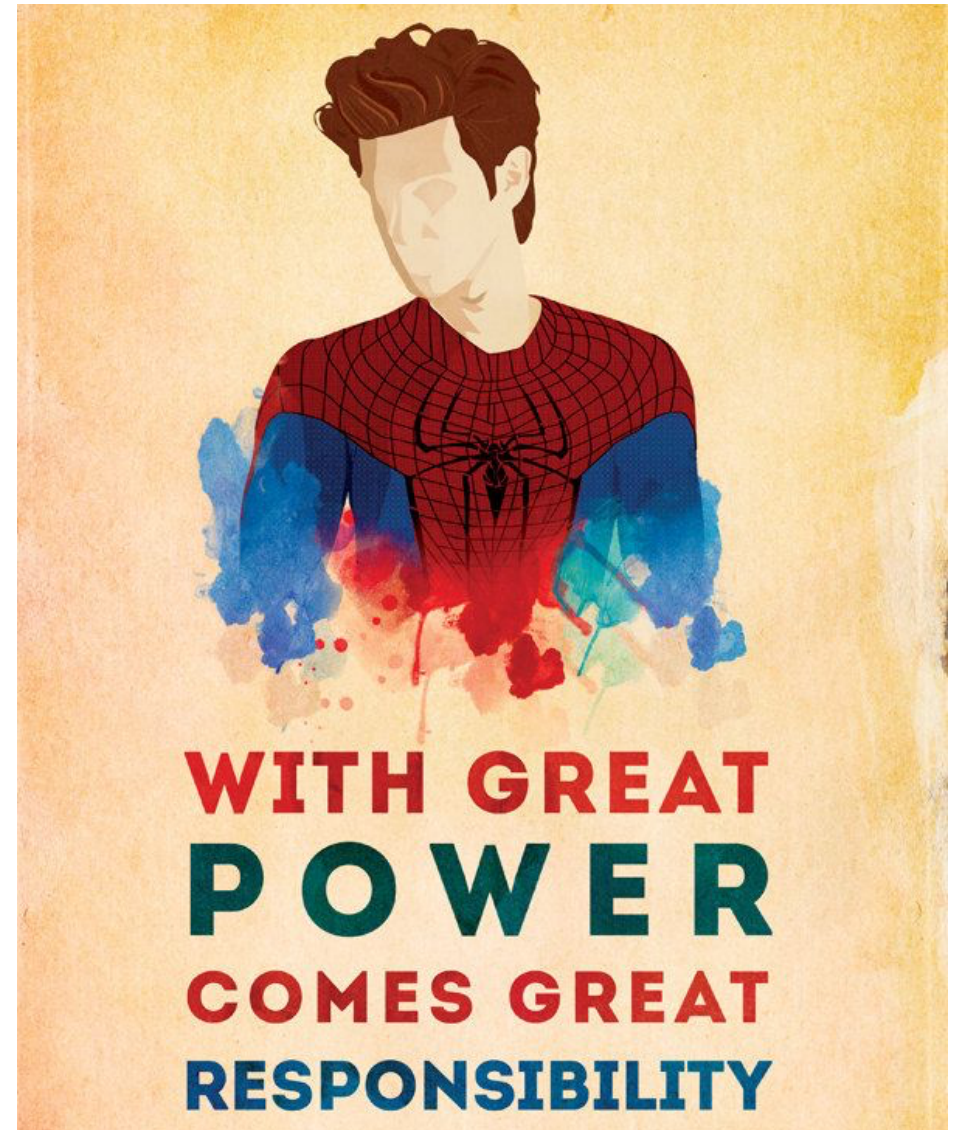
Coordinate the team

Validate merge requests

Manage continuous integration

Leaders Bonus

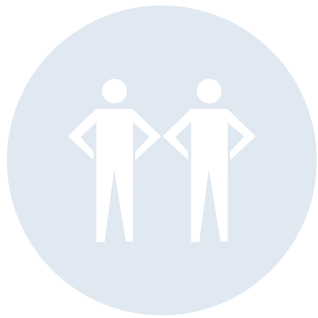
+0-15%



Leader
Bonus



Develop!

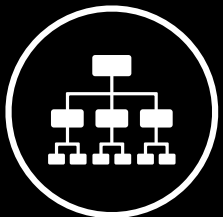


Follow issues assigned by leaders
during a sprint

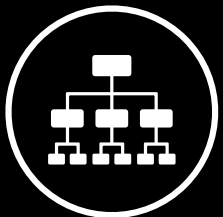
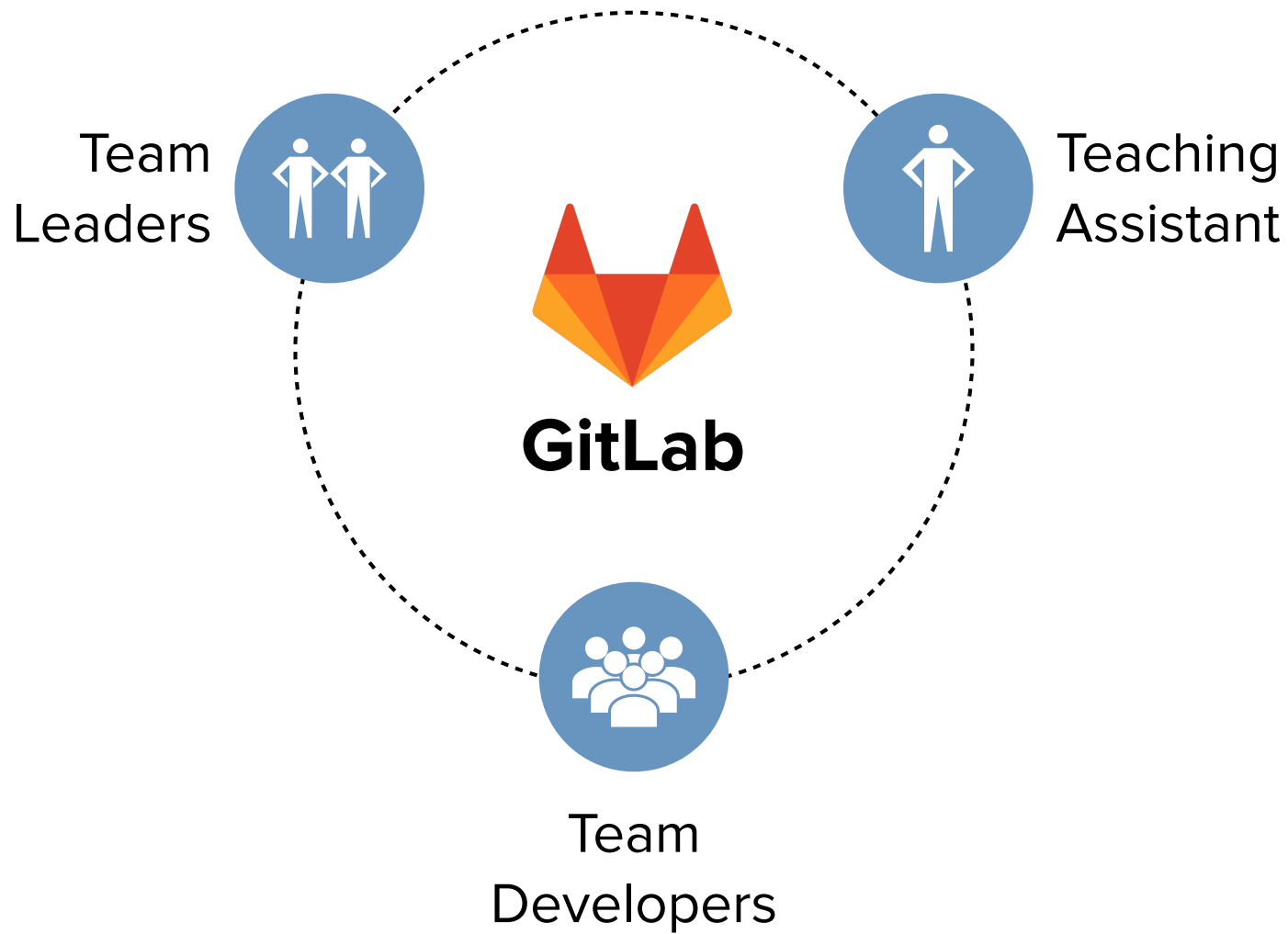


Team
Developers

Please note, **team leaders are
also developers!**



Project
Organization

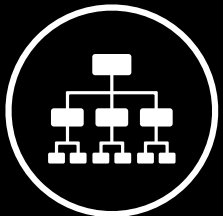


Project
Organization

Team

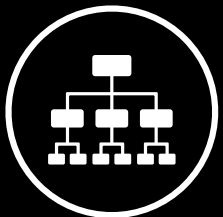
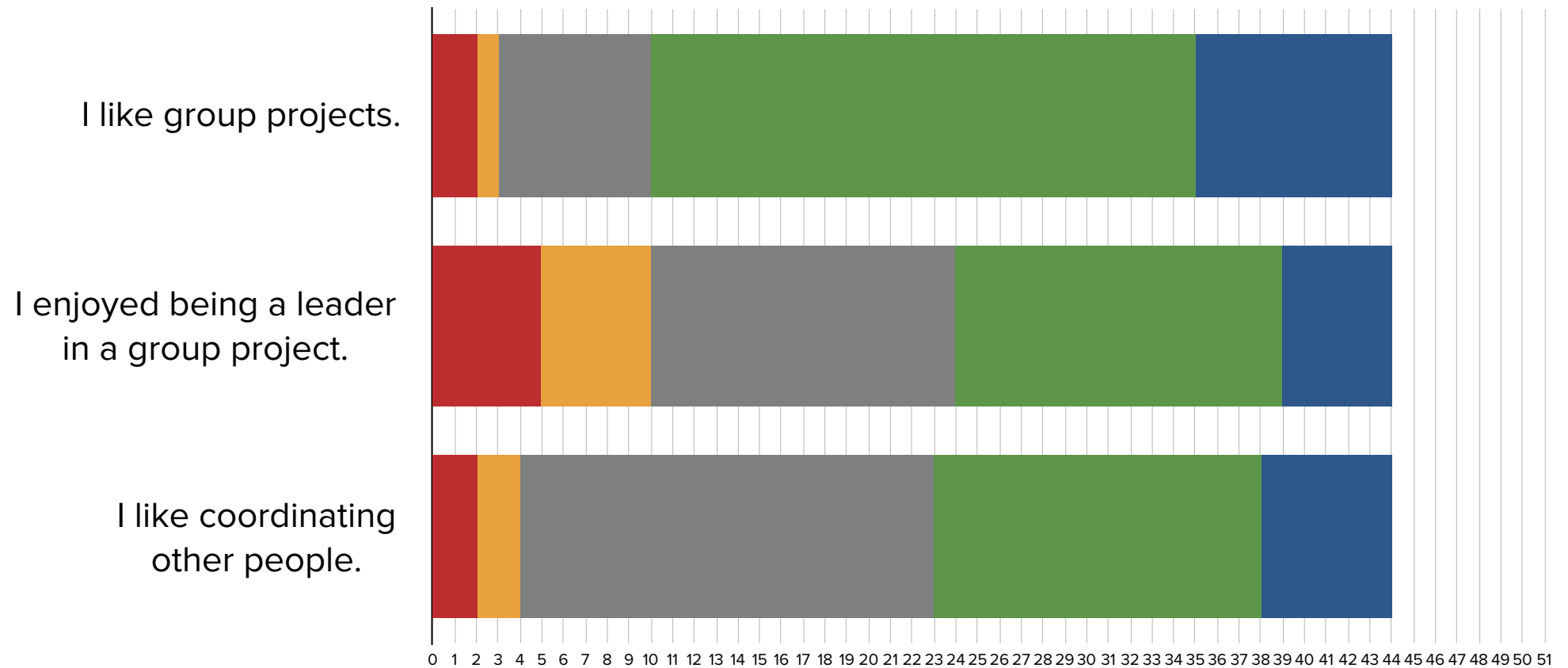
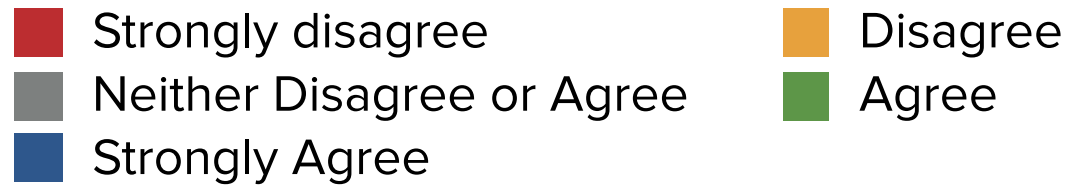


Backend

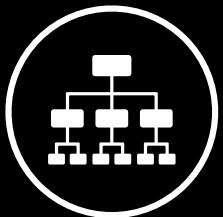
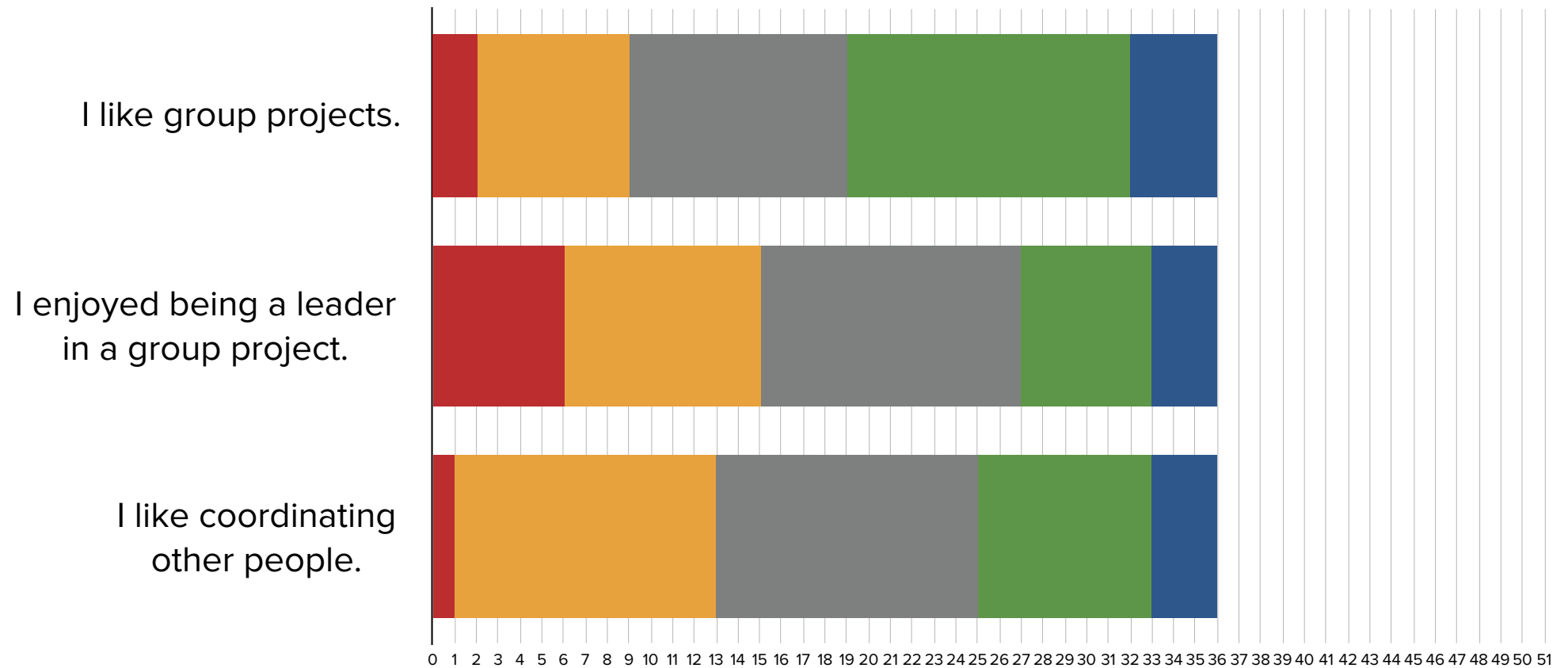
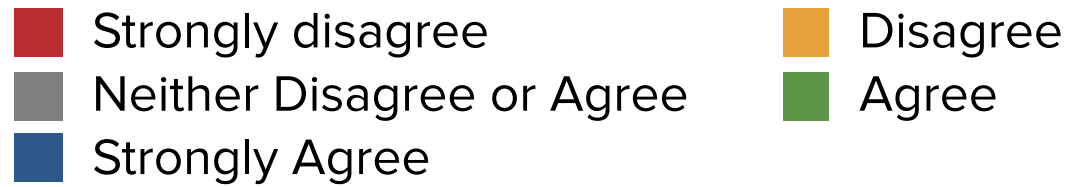


**Project
Organization**

Team Building



Survey (2024)



Survey (2025)

Leaders Responsibilities

In addition to **development**:

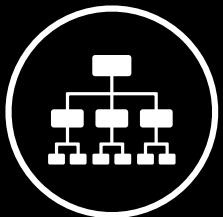
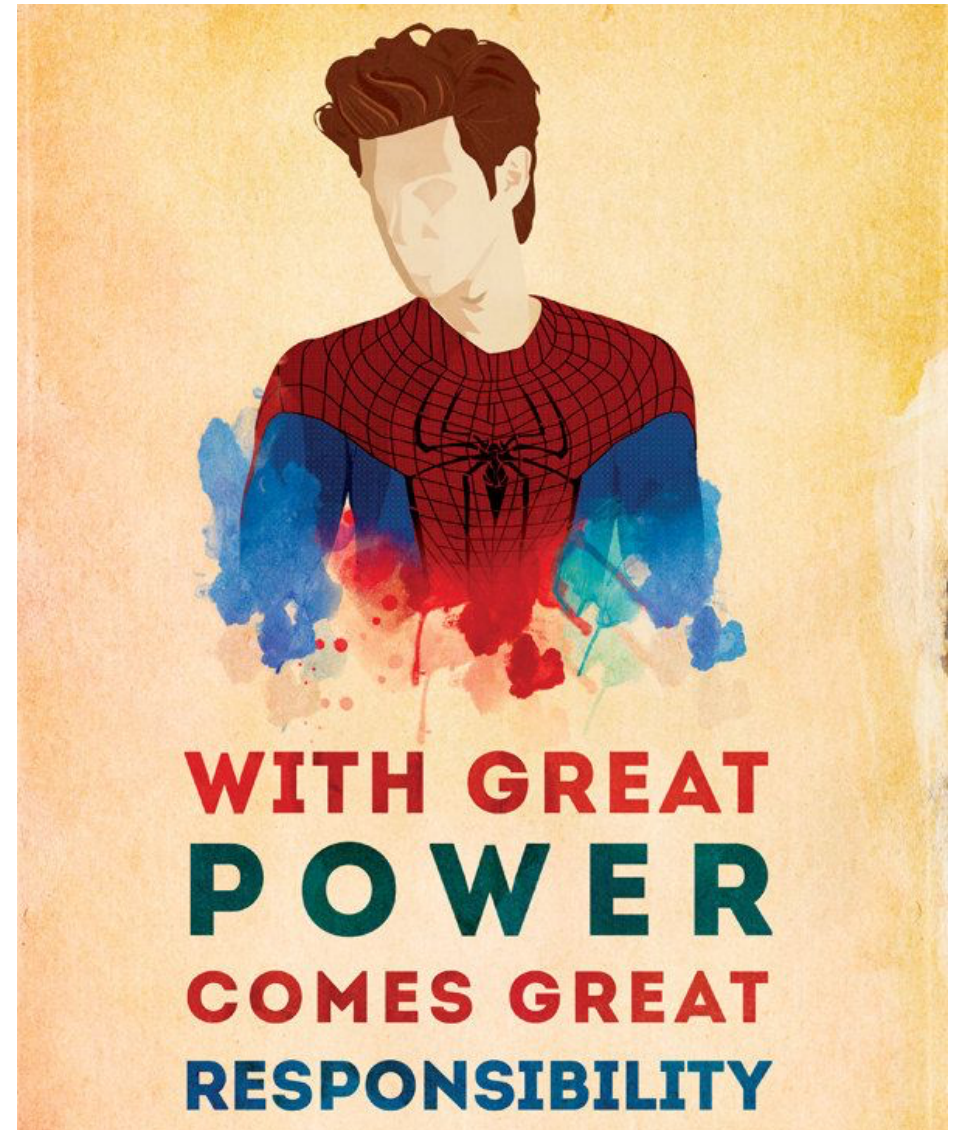
Coordinate the team

Validate merge requests

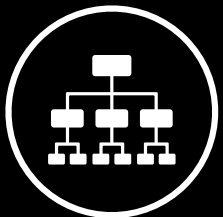
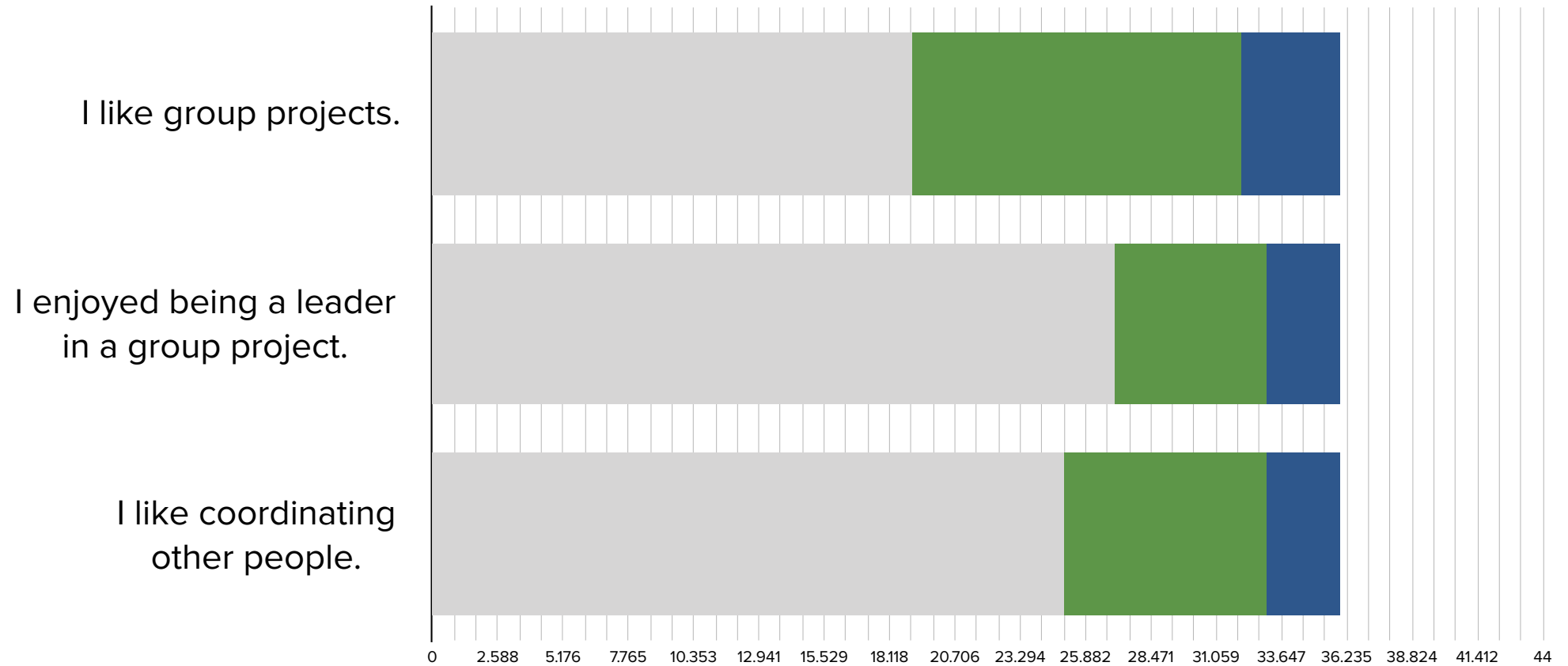
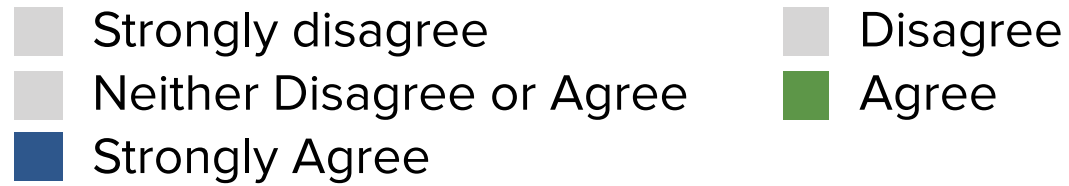
Manage continuous integration

Leaders Bonus

+0-15%



Leader
Bonus



Development Requirements

Base:
Stay Agile:
lightweight & iterative
Implement CI
Follow Lab01 guidelines (GitLab)
Pair Programming

For the Rest:
Pick your Own Process

Daily Meetings?
TDD?
Sprints (SCRUM) or Iterations (XP)?
...

Do Not Squash Commits
(HUGE penalty)
Commit VERY frequently (Hourly) -
even if you break pipelines in your
issue branch



Agile
Process

**If you decide to do
daily meetings,
you can invite a TA (or the instructor).**

**At any time, you can ask for help
or clarifications to the instructor
or TAs.**

**During course times, we'll answer
pretty much promptly or schedule
meetings.
Outside, there might be delay.**



Agile
Process

**All communication should
happen via Gitlab or Teams**

**We're gonna set up
dedicated team channels on Teams**

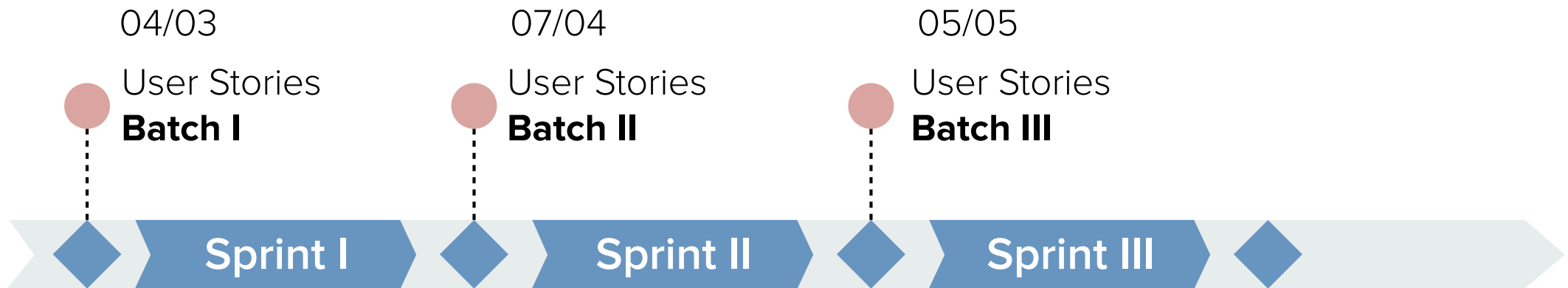
**NO SOCIAL MEDIA TO
EXCHANGE CODE**

**Exception on some platforms (Discord)
may be discussed with team leaders
but we prefer to be able
to lurk on your discussions**



Communication

(Rough) Timeline



You should work approximately 2 hours per (week) day during the course hours if there is no lecture, + 1 to 2 hours per (week)day outside course hours.



Project
Timeline

Grading

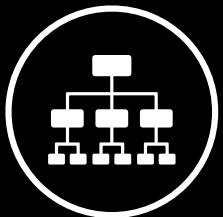
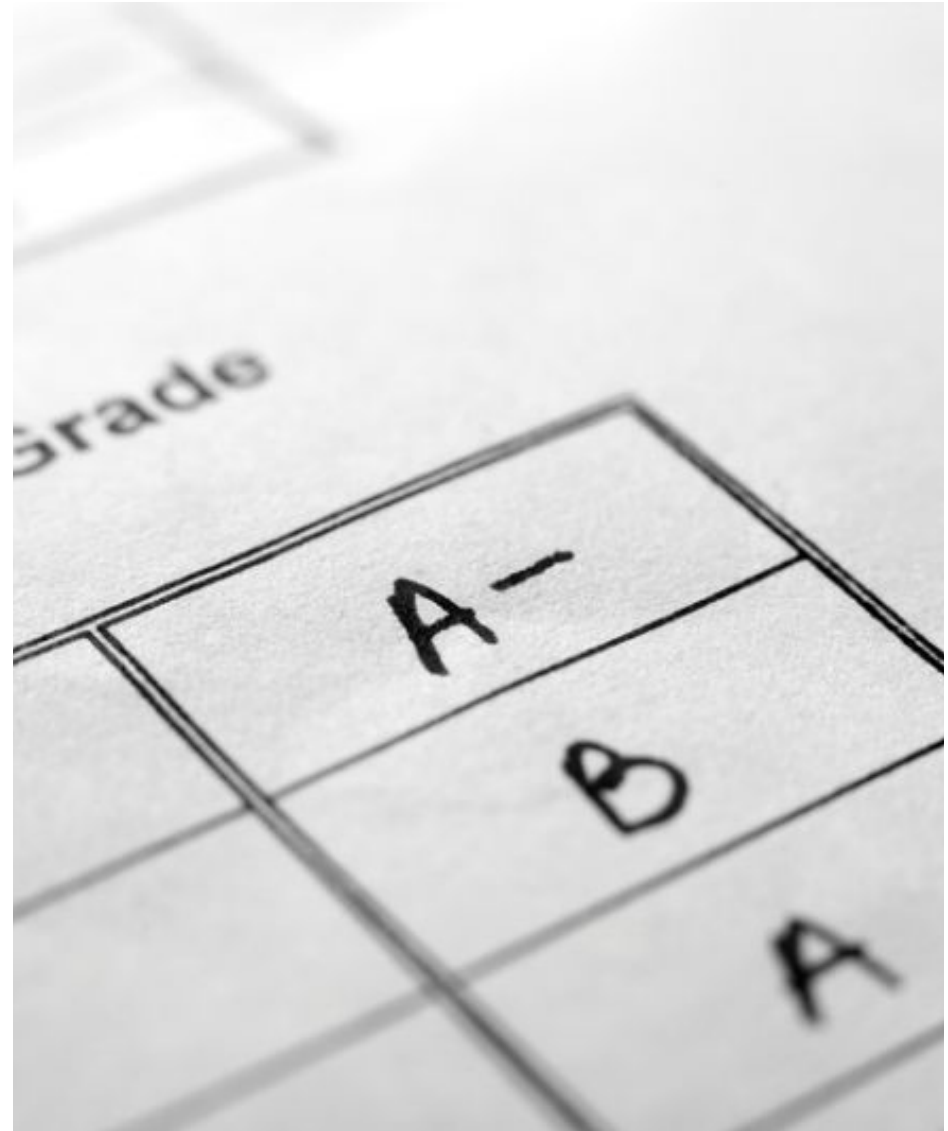
Grading Scheme

Each user story is worth **n points**

Stories can be validated
at any **Friday session**.

Exceptions may happen.

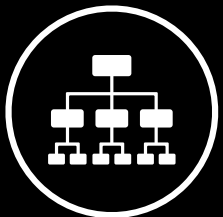
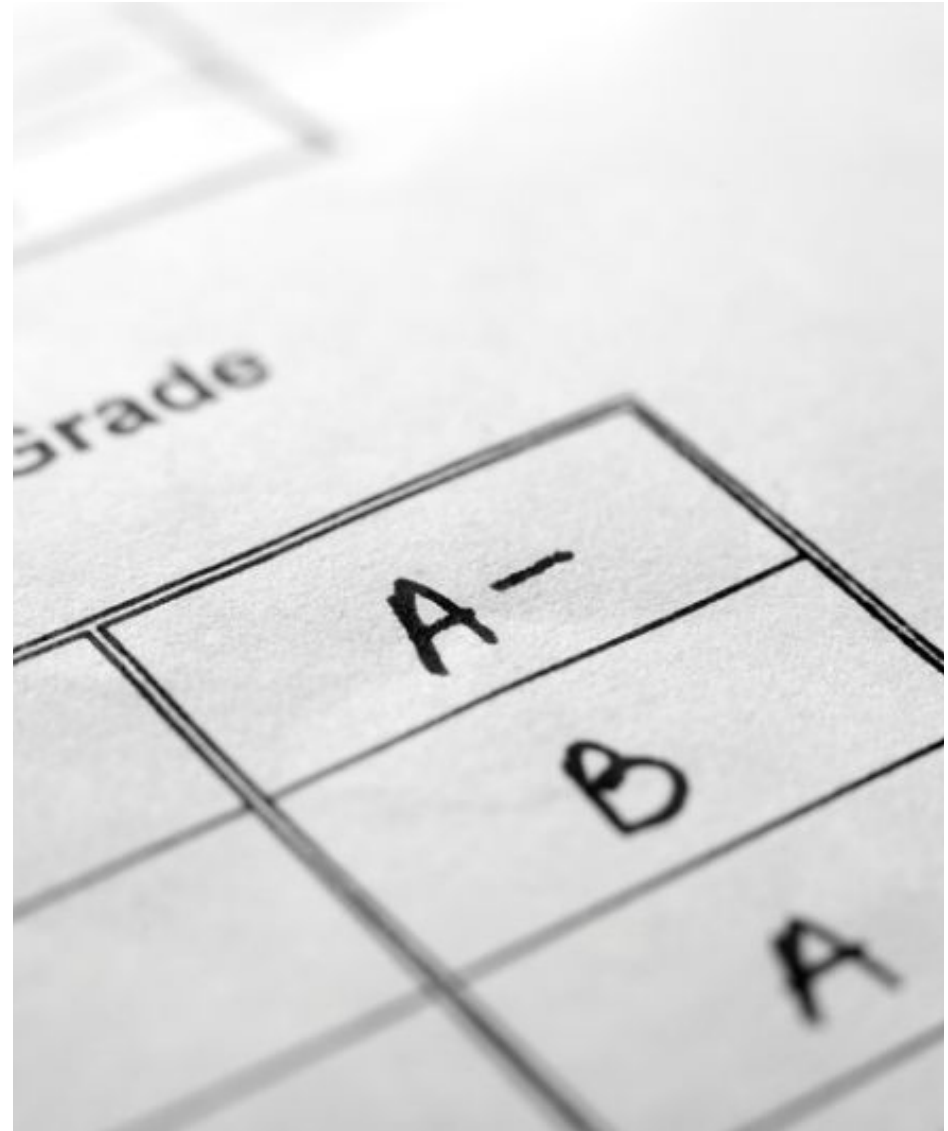
The validation might be
worth **m points** ($m \leq n$) as judged
by Instructors & TAs, acting as
customers & possibly code
reviewers



Project
Grading

Grading Scheme

If you are not satisfied with the grade of a story, you can fix it and re-validate it in the future but **with a penalty of 20% of points.**

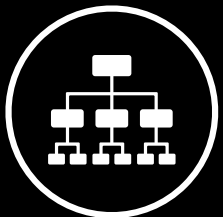
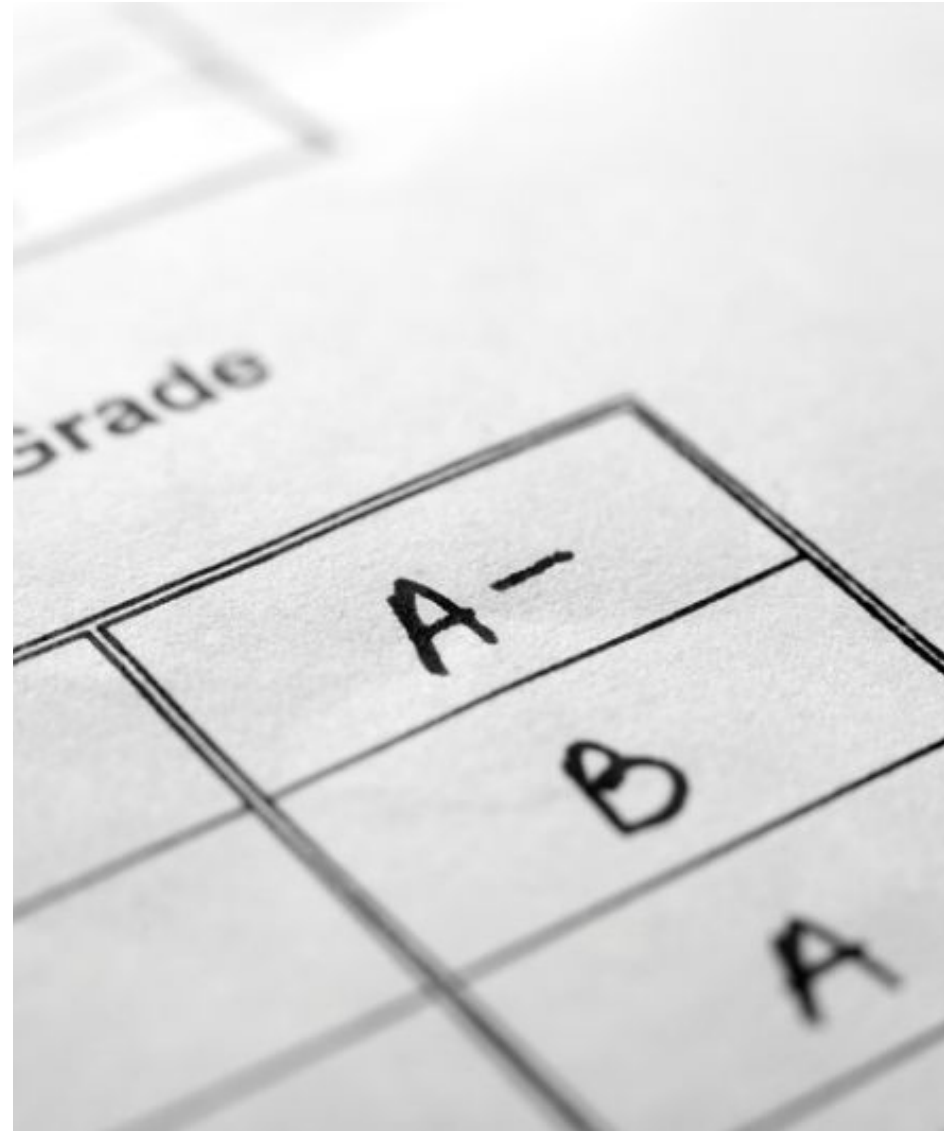


Project
Grading

Grading Scheme

40% of the grade for the project is determined by the grade of user stories **for the whole team**

The rest is a **personal** grade (60%).

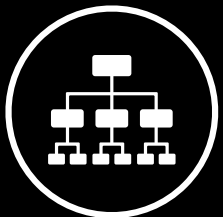
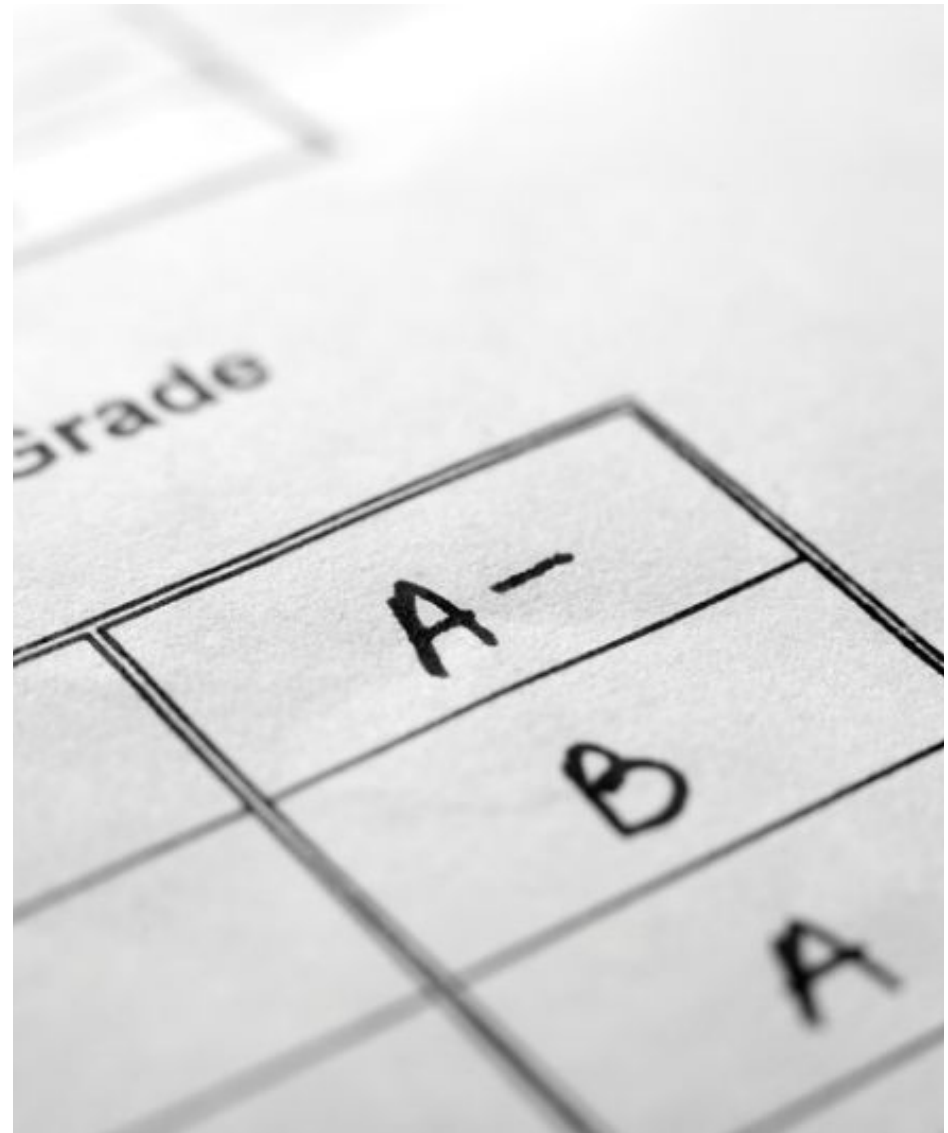


Project
Grading

Grading Scheme

Each user story must have a **list of people** who worked on the corresponding issue(s), declared by the team leaders.

The list may include **up to 50% of the team**, but we're going to check that such people have substantially worked on it.



Personal
Grade

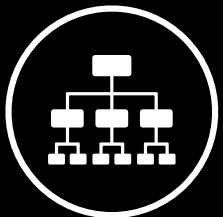
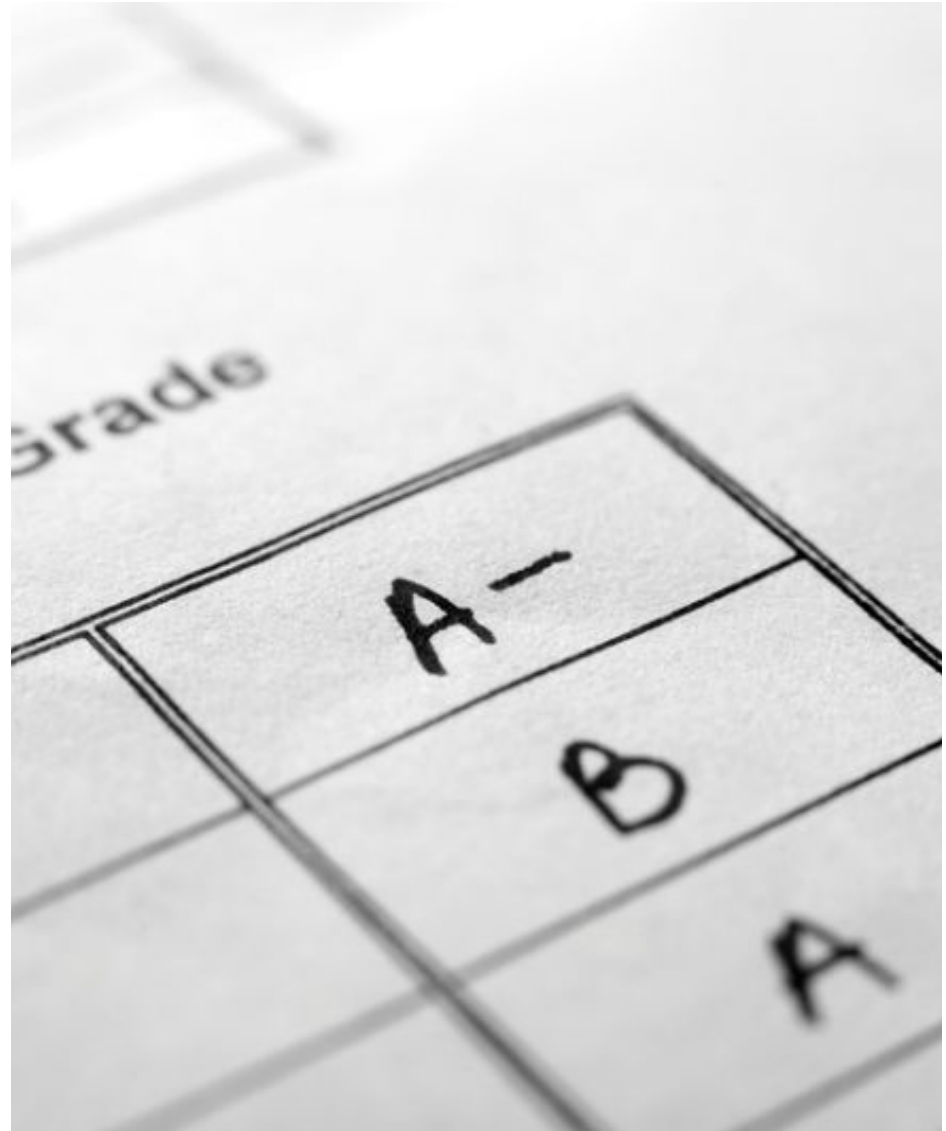
Suggestion

Assign a issue to a user story

Split it into different parts

Assign them to people by
mentioning them on gitlab

Frequently integrate

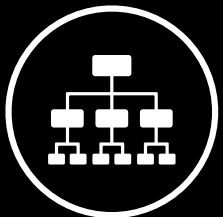
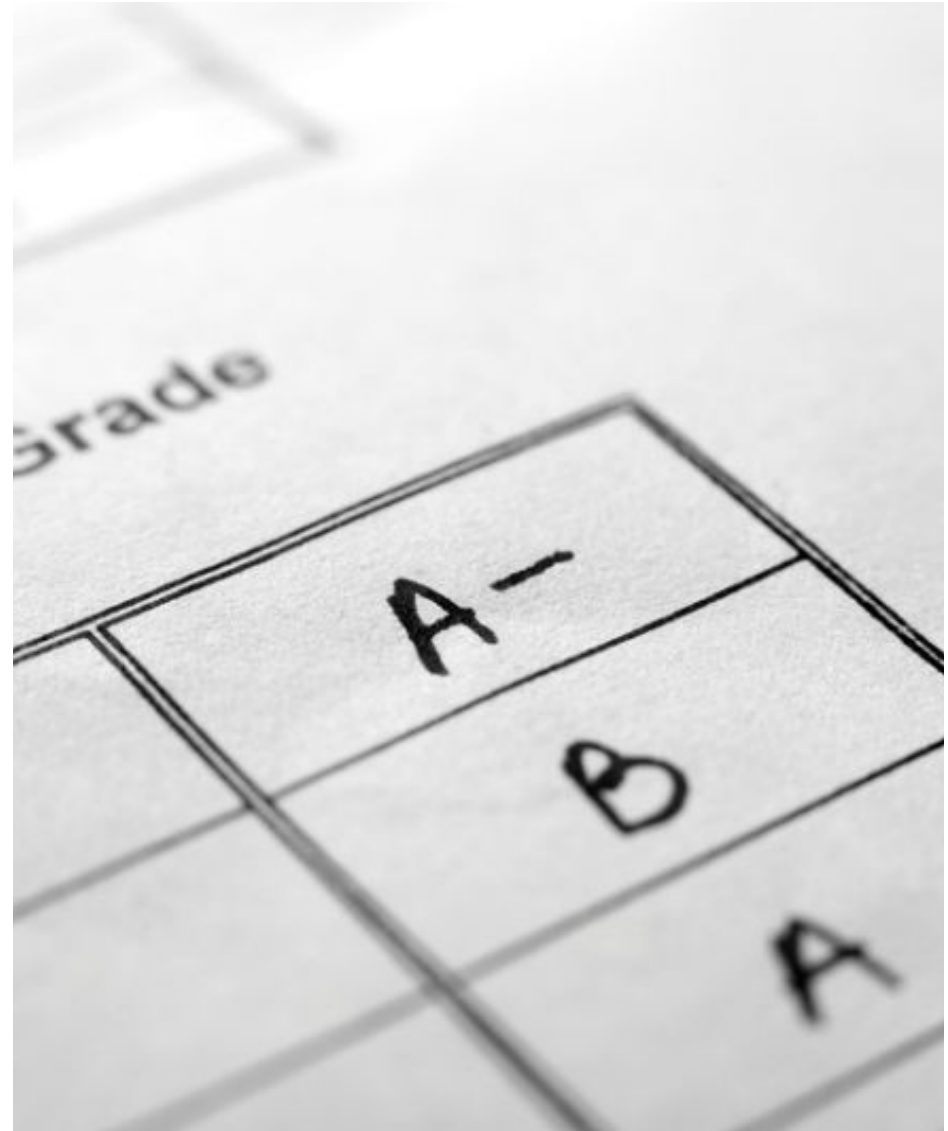


Personal
Grade

Grading Scheme

The personal grade will depend on the grades of the **issues** you worked on.

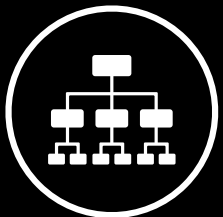
Penalties will be added if you didn't really work substantially (e.g., you **disappeared** and stopped working at all for long periods of time).



Personal
Grade

Validations

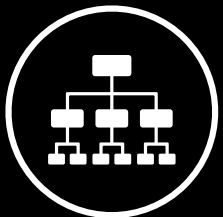
When you have at least 3 user stories ready to validate, the leaders ask for an appointment for validation during course time. All Team members must participate.



Project
Grading

DISCLAIMER

Penalties at the individual level
are defined by the **accountability**
of the person
(e.g, issues completed, commits,
gitlab data)

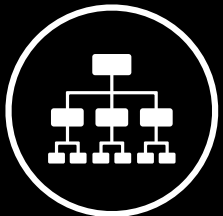


Project
Grading

EXAMPLE

You stop working or disappear
from the course:

COURSE FAILED

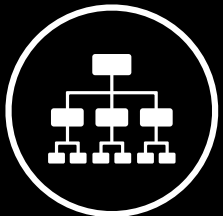


Project
Grading

EXAMPLE

As a team leader I do all the work
and don't favor teammates
cooperation:

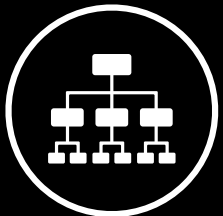
PENALTY



NO EXCUSES LIKE

We always worked on his/her
machine

Pair Programming is Mandatory
Switch Roles and
Commit Frequently
Use Co-Authored-By (see notes)



Project
Grading

That's all!